Sumon Datta, Ph.D. E.I.

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Director, Research in Irrigation Scheduling & Engineering (RISE) Laboratory
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Links to Online Resources

- Personal Website: https://sumondatta.github.io/rise-lab/
- Operational Agriculture and Soil Information System – OASIS Tool: OASIS Tool
- LinkedIn: LinkedIn Profile
- OSU Experts Link: OSU Experts Profile
- OrcID: ORCID Profile

Career Summary

Grantsmanship Summary

- Secured **26** grants totaling **$54.39M**, with approximately **$2.99M** directly awarded to my program.
  - OSU: **23** grants totaling **$53.34M**, with approximately **$2.51M** directly awarded to my program, since 2023.

Research & Extension Summary

- Produced **24** peer-reviewed publications:
  - **19** peer-reviewed journal papers & conference proceedings.
  - **5** peer-reviewed Extension factsheets.
- Delivered **72** presentations: **33** conference presentations orally, & **22** conference presentations through posters, along with **17** Extension presentations and invited talks.

Teaching & Mentoring Summary

- Developed/Taught **3** distinct courses, for **5** times.
- Delivered **8** guest lectures and participated in **6** student recruitment events.

Tools Developed

- Developed OASIS Tool for OK, KS, and TX farmers & policymakers.

Service & Recognition

- Contributed to **30** professional service activities:
  - **12** journal reviewer appointments, **1** editorial board role, and **1** proposal review.
  - **3** professional memberships and **13** committee participations.
- Accumulated **17** recognitions: **6** student awards & **11** personal awards.

Education & Employment Background

Education

Ph.D. in Biosystems Engineering 5/2020
Department of Biosystems & Agricultural Engineering
Oklahoma State University, Stillwater, OK
Major Advisor: Dr. Saleh Taghvaeian
Dissertation: Measurement and Modeling of Soil Moisture for Irrigation Management

M.S. in Agricultural & Biosystems Engineering 12/2016
Department of Agricultural & Biosystems Engineering
North Dakota State University, Fargo, ND
Major Advisor: Dr. Shafiqur Rahman
Thesis: Quantification and Characterization of Particulate Matter Generated from Unpaved Roads in the Oil Development Area of Western North Dakota

B.S. in Agricultural Engineering 1/2014
Agricultural Engineering & Technology
Bangladesh Agricultural University, BD
Major Advisor: Md. Siddikur Rahman
Project: Waste Generation and Management Practices in BSCIC, Mymensingh

Professional Positions Held

Assistant Professor & Irrigation Engineer 7/2023 – Present
75% Research, 25% Teaching
Biosystems & Agricultural Engineering, Oklahoma State University, Stillwater, OK

Assistant Extension Professor & Extension Agricultural Engineer 5/2022 – 6/2023
100% Extension
Cooperative Extension, University of Maine, Orono, ME

Postdoctoral Fellow 6/2020 – 5/2022
Biosystems & Agricultural Engineering, Oklahoma State University, Stillwater, OK

Graduate Research Assistant 1/2017 – 5/2020
Biosystems & Agricultural Engineering, Oklahoma State University, Stillwater, OK

Graduate Research Assistant 1/2015 – 12/2016
Agricultural & Biosystems Engineering, North Dakota State University, Fargo, ND

Undergraduate Research Assistant 1/2013 – 10/2014
Agricultural Engineering & Technology, Bangladesh Agricultural University, BD

Certifications

Engineer Intern (EI), Maine, License: # EI8079

Grantsmanship

Overview

Total $ in Grants Awarded

Total: **$54.39M** (**$53.34M** at OSU since 2023).
As Lead PI: **$1.91M** (**$913.3K** at OSU since 2023).
As Co-PI: **$52.47M** (**$52.43M** at OSU since 2023).

My Share in $

Total: ∼**$2.99M** (∼**$2.51M** at OSU since 2023).
As Lead PI: **$1.33M** (**$852.3K** at OSU since 2023).
As Co-PI: ∼**$1.66M** (∼**$1.65M** at OSU since 2023).

Grants Awarded

At Oklahoma State University (Current Position as a Faculty, 7/2023 - Present)

1. Datta et al. (2026). Sustainable Intensification of Targeted Agricultural Systems Innovation Lab (SITAS-IL). US Department of State Feed the Future Innovation Lab. Co-PI. 10/2026 – 9/2031. F&A[a]: 49.6%. $39,999,752[b]. (∼$250,000[c])
2. Datta et. al. (2026). Improving Water Use Efficiency of Mature Pecan Trees for Small and Medium-Sized Farms. USDA NIFA AFRI Foundational and Applied Sciences Program. Co-PI. 11/2026 – 11/2029. F&A: 42.57%. $590,735. ($295,000)
3. Datta, S. (2026). Developing Scientific Variable Irrigation Strategies to Increase Water Use Efficiency of Sorghum in Oklahoma. Oklahoma Sorghum Checkoff. Lead PI. 10/2026 – 9/2029. F&A: 0%. $60,000. ($60,000)
4. Datta et. al. (2026). Integrating In-field Soil Moisture and Canopy Temperature Data to Improve Cotton Irrigation Efficiency and Water Productivity in the Southwestern US. USDA NIFA AFRI Foundational and Applied Sciences Program. Lead PI. 9/2026 – 9/2029. F&A: 42.57%. $588,000. ($527,000)
5. Datta et al. (2026). Regional Forecasting of Evapotranspiration and Crop Water Demand Using Open ET, GraphCast Weather Predictions, and NISAR Soil Moisture Observations. NASA A8 Water Resources Program. Co-PI. 6/2026 – 5/2029. F&A: 49.6%. $900,000. ($234,825).
6. Datta et al. (2026). Scaling a Data-Driven Decision-Support Platform for Irrigation Scheduling to Improve Water Conservation in the Ogallala Aquifer Region. Ogallala Aquifer Program. Co-PI. 1/2027 – 12/2028. F&A: 0%. $79,898. ($79,898)
7. Datta et al., (2026). Producer Attitudes Towards Groundwater Conservation Policies and Programs in the High Plains Aquifer Region. USDA NIFA AFRI Foundational and Applied Sciences. Co-PI. 5/2026 – 4/2029. F&A: 42.85%.$649,946 ($12,857)
8. Datta, S. (2026). Deployment and Testing of Oklahoma Agricultural Scientific Irrigation Scheduler – OASIS Tool. Oklahoma Mesonet. Lead PI. 10/2025 – 12/2027. F&A: 0%. $41,940 ($41,940)
9. Datta, S. (2026). Data-driven Canopy Temperature Modeling for Irrigated Cotton: Phase III – Field Testing & OASIS Tool Integration. Cotton Incorporated Core Research Program. Lead PI. 1/2026 – 12/2026. F&A: 15%. $20,000 ($20,000)
10. Datta, S. (2026). Field-scale Sensor-informed Water Allocation for Sustainable Cotton Production in Lugert-Altus Irrigation District. Cotton Incorporated – Oklahoma State Support Committee. Lead PI. 1/2026 – 12/2026. F&A: 0%. $14,200 ($14,200)
11. Datta, S. (2026). Expanding OASIS Tool Kansas Farmers. Kansas Water Institute at Kansas State University. Lead PI. 1/2026 – 12/2027. F&A: 0%. $46,094 ($46,094)
12. Datta, S. (2026). TxSIS: TX Scientific Irrigation Scheduler - Expanding OASIS Tool for West TX Farmers. Davis College Water Center at Texas Tech University. Lead PI. 1/2026 – 12/2027. F&A: 0%. $46,094 ($46,094)
13. Datta et al. (2025). Water Meters for Smart Savings and Conservation? Examining Water Meters as a Tool for Sustainable Irrigation in Oklahoma's Rural Communities. Rural Renewal Initiative. Co-PI. 1/2025 – 12/2027. F&A: 0%. $90,000 ($8,000)

[a] Facilities & Administration cost
[b] Total $ awarded for this grant
[c] Number inside the parentheses indicates $ awarded to my program

14. Datta et al. (2025). Cost-effective Edge Sensors: Resourcing AI for Irrigation Networks (CES-RAIN). OSU College of Engineering, Architecture, and Technology ERSF Seed Grant. Co-PI. 8/2025 – 7/2026. F&A: 0%. $25,000 ($0)
15. Datta et al. (2025). Sustainable Irrigation and Climate Adaptation in Southern High Plains: A Satellite-Enabled and Peer-Led Model. USDA NIFA AFRI Sustainable Agricultural Systems. Co-PI. 2/2025 – 1/2030. F&A: 42.85%. $10,000,000 ($744,897)
16. Datta, S. (2025). Deployment of Advanced Autonomous Oklahoma Irrigation Planner. Oklahoma Mesonet Products Development Fund. Lead PI. 1/2025 – 9/2025. F&A: 0%. $18,000 ($18,000)
17. Datta, S. (2025). Data-driven Canopy Temperature Modeling for Irrigated Cotton: Phase II – Evaluation of Instrumentation Requirement for Data-driven Approaches. Cotton Incorporated Core Funding. Lead PI. 1/2025 – 12/2025. F&A: 15%. $20,000 ($20,000)
18. Datta, S. (2024). Investigating Changes in Consumptive Water Use in Furrow and Subsurface Drip Irrigated Cotton Fields. USGS 104b. Lead PI. 9/2024 – 8/2025. F&A: 49.6%. $24,995 ($24,995)
19. Datta, S. (2024). Data-driven Canopy Temperature Modeling for Irrigated Cotton: Phase I – Evaluation of State-of-the-art Data-driven Approaches. Cotton Incorporated Core Funding. Lead PI. 6/2024 – 12/2024. F&A: 15%. $13,998 ($13,998)
20. Datta, S. (2024). Building an Advanced Autonomous Oklahoma Irrigation Planner. Oklahoma Mesonet Products Development Fund at Oklahoma State University. Lead PI. 5/2024 – 12/2024. F&A: 0%. $18,000 ($18,000)
21. Datta et al. (2024). Oklahoma Master Irrigator Program and Ogallala Aquifer Study. Oklahoma Water Resources Board/Oklahoma Conservation Commission. Co-PI. 1/2022 – 8/2024. $60,081 F&A: 0%. ($19,734)
22. Datta, S. (2024). Comparison of Consumptive Water Use of Different Irrigation Systems in Lugert-Altus Irrigation District. 2024 OSU Buchanan Undergraduate Research Fund. Lead PI. 1/2024 – 5/2024. F&A: 0%. $2,000 ($2,000)
23. Datta et al. (2024). Annual Funding for Conducting Irrigation System Audits for Oklahoma Master Irrigator Program and Mobile Irrigation Lab. Oklahoma Conservation Commission/Oklahoma Water Resources Board. Co-PI. 1/2024 - 12/2024. F&A: 0%. $31,622 ($7,866)

At University of Maine (As a Faculty, 5/2022 – 6/2023)

24. Datta et al. (2023). Piloting Innovation in Wild Blueberry: Team WILD – Wild Blueberry Innovations Led by Data. Maine Technology Institute. Lead PI. 5/2023 – 12/2026. F&A: 19%. $1,000,000 ($473,836)
25. Datta et al. (2023). Socio-Economic and Environmental Analyses for Using Woody Biochar to Conserve Water and Sustain Agriculture in Maine. University of Maine Water Resources Research Institute. Co-PI. 9/2023 – 8/2024. F&A: 47.5%. $40,000 ($2,000)

At Oklahoma State University (As a Student, 1/2017 – 5/2020)

26. Datta et al. (2018). Conserving Agricultural Water Resources Using Smart Technologies. USGS 104b. Co-PI. 9/2018 – 8/2019. F&A: 49.6%. $5,000 ($5,000)

Research & Extension Activities

Overview

My research objectives are to monitor, study, and model the bio-physical interactions within cropping systems boundary, particularly irrigated ones, to find answers to two most important questions: when should we turn on and off watering – also known as scientific irrigation scheduling. I utilize in-situ sensors (e.g., soil moisture and canopy temperature sensors), satellite products (e.g., high resolution imageries), models (e.g.,

soil water balance, numerical models), and recently, data-driven approaches (e.g., machine learning/deep learning) to find best and operationally feasible approaches to scientific irrigation scheduling. My Extension efforts are focused on transferring and communicating scientific knowledge to stakeholders such as agricultural producers and state agencies facing issues with irrigation water and its scheduling on both field and regional scales; to policymakers to help with formulating water allocation strategies for the state and beyond; and to scientific community to contribute to existing and new knowledge on irrigation management through application-based, stakeholder-driven integrated Research and Extension projects.

Google Scholar Metrics

- Total number of citations: 570 (as of October 1, 2026)
- h-index: 9
- i10-index: 8

Peer-reviewed Journal Publications & Conference Proceedings

1. Meadi, S., Kalayambakam Janardhan, A. Y., **Datta, S.**, & Das, H. (2026). Canopy Temperature Sensor Data Optimization for Irrigation Scheduling. 2026 IEEE DCAS Conference, Dallas, TX, April 10-12, 2026. *DOI to be available in June, 2026.*
2. Kalayambakam Janardhan, A. Y., **Datta, S.**, Crick, C., Taghvaeian, S., & Kundu, P. K. (2026). Data-driven Modeling of Soil Moisture at Irrigated Fields. 2026 IEEE DCAS Conference, Dallas, TX, April 10-12, 2026. *DOI to be available in June, 2026.*
3. DeJonge, K. C., Allen, R. G., Kilic, A., Thorp, K. R., Kukal, M. S., Marek, G., Altenhofen, J., Blankenau, P., **Datta, S.**, Grabow, G., Hashem, A. A., Kisekka, I., Kjaersgaard, J., Marek, T., Peters, T., Porter, D., Reba, M., Rudnick, D., Senay, G., Sharma, V., Sridhar, V., Sun, G., Taghvaeian, S., Trezza, R., & Trout, T. (2025). Evapotranspiration Terminology and Definitions. ASCE Journal of Irrigation and Drainage Engineering, 151(5): 06025003. https://doi.org/10.1061/jidedh.ireng-10491
4. **Datta, S.**, Taghvaeian, S., Sibley, M., Gholson, D. M., Yost, M., Long, M. A., Bali, K. M., Zaccaria, D., Davis-Conger, S. L., & Ritchie, L. A. (2025). Irrigation and Water Conservation Practices of Surface-Irrigated Croplands in West and South Regions of the US. Journal of the ASABE, 68(3), 503-511. https://doi.org/10.13031/ja.16257
5. Akbar, M. U., Mirchi, A., Arshad, A., Alian, S., Mehata, M., Taghvaeian, S., Khodkar, K., Kettner, J., **Datta, S.**, & Wagner, K. (2025). Multi-model Ensemble Mapping of Irrigated Areas Using Remote Sensing, Machine Learning, and Ground Truth Data. Agricultural Water Management, 312, 109416. https://doi.org/10.1016/j.agwat.2025.109416
6. Kettner, J., Taghvaeian, S., Pokhrel, N., Mirchi, A., Zhang, L., Mehata, M., & **Datta, S.** (2025). Water Dynamics of Variably Irrigated Pecan Trees in Oklahoma. Journal of the American Pomological Society, 78(2): 85-99. https://doi.org/10.71318/tak32v50
7. Munmun, T. H., Islam, M. T., Rahman, M. M., Islam, M. A., **Datta, S.**, Das, N., Akter, J., & Adham, A. K. M. (2024). Rice Cultivation Under Raised Bed Conserving Irrigation Technique: Effects of Bed Width on Soil Wetness and Yield. Paddy and Water Environment. https://doi.org/10.1007/s10333-023-00957-3
8. **Datta, S.**, & Taghvaeian, S. (2023). Soil Water Sensors for Irrigation Scheduling in the United States: A Systematic Review of Literature. Agricultural Water Management, 278, 108148. https://doi.org/10.1016/j.agwat.2023.108148
9. Mehata, M., **Datta, S.**, Taghvaeian, S., Mirchi, A., & Moriasi, D. N. (2023). Effects of Soil Data Accuracy on Outputs of Irrigation Scheduling Tools. Journal of the ASABE, 66(3), 677-687. https://doi.org/10.13031/ja.15323
10. Mehata, M., **Datta, S.**, Taghvaeian, S., Ochsner, T., Mirchi, A., & Moriasi, D. N. (2023). Performance

Student advisees are underlined.

of a Multi-sensor Capacitance Probe in Estimating Soil Water Content and Field Capacity. Journal of the ASABE, 66(2), 253-261. https://doi.org/10.13031/ja.15416

11. Mehata, M., **Datta, S.**, Taghvaeian, S., Mirchi, A., Moriasi, D., & Starks, P. J. (2022). Simulating Soil Water Content of Irrigated Fields: The Effects of Variable Soil Data and Root Water Uptake Distribution. Journal of the ASABE, 65(3), 587-597. https://doi.org/10.13031/ja.14856
12. **Datta, S.**, Mehata, M., Taghvaeian, S., Moriasi, D., & Starks, P. J. (2021). Quantifying Water Fluxes of Irrigated Fields in an Agricultural Watershed in Oklahoma. ASCE Journal of Irrigation and Drainage Engineering, 147(7): 04021026. https://doi.org/10.1061/(ASCE)IR.1943-4774.0001570
13. Ajaz, A., **Datta, S.**, & Stoodley, S. (2020). High Plains Aquifer–State of Affairs of Irrigated Agriculture and Role of Irrigation in the Sustainability Paradigm. Sustainability, 12(9): 3714. https://doi.org/10.3390/su12093714
14. **Datta, S.**, & Ajaz, A. (2019). Geospatial Data Assimilation and Mapping Groundwater Vulnerability in High Plains Aquifer using DRASTIC Model. Fundamental and Applied Agriculture, 4(3): 933-942. http://dx.doi.org/10.5455/faa.53506
15. Rana, M. M., Islam, M. T., **Datta, S.**, Rahman, M. M., & Adham, A. K. M. (2019). Suitability of Powerplant Disposed Water for Irrigation of Ashuganj Agro-Irrigation Project in Bangladesh. Progressive Agriculture, 30(1), 113-124. https://doi.org/10.3329/pa.v30i1.42218
16. **Datta, S.**, Rahman, S., Borhan, M.S., Saini-Eidukat, B., Cihacek, L., & Ringwall, K. (2019). Quantification and Characterization of Particulate Matter Generated from Unpaved Roads in the Oil Development Area of Western North Dakota. Transactions of the ASABE, 62(3): 615-625. https://doi.org/10.13031/trans.13169
17. Masasi, B., Taghvaeian, S., Boman, R., & **Datta, S.** (2019). Impacts of Irrigation Termination on Cotton Yield and Irrigation Requirement. Agriculture, 9(2), 39. https://doi.org/10.3390/agriculture9020039
18. **Datta, S.**, Taghvaeian, S., Ochsner, T., Moriasi, D., Gowda, P., & Steiner, J. (2018). Performance Assessment of Five Different Soil Moisture Sensors under Irrigated Field Conditions in Oklahoma. Sensors, 18(11), 3786. https://doi.org/10.3390/s18113786
19. Rahman, S., **Datta, S.**, & Islam, S. (2014). Waste Generation and Management Practices in BSCIC, Mymensingh. Journal of Environmental Science and Natural Resources, 7(1): 47-51. https://doi.org/10.3329/jesnr.v7i1.22143

Peer-reviewed Extension Factsheets Published

1. Sharma, S., Wagner, K., & **Datta, S.** (2025). Using Soil Moisture Trend Values from Moisture Sensors for Irrigation Decisions. Oklahoma Cooperative Extension Service, Accepted & Pending Factsheet Number Assignment.
2. Krueger, E. S., Ashrafi, A., Ochsner, T., & **Datta, S.** (2025). Determining Field Capacity Using Continuous Soil Water Content Data. Oklahoma Cooperative Extension Service, PSS-2403.
3. **Datta, S.**, Kettner, J., Mehata, M., & Taghvaeian, S. (2025). Irrigated Agriculture in Oklahoma. Oklahoma Cooperative Extension Service, BAE-1530.
4. Frazier, S., & **Datta, S.** (2024). Understanding the Results of Master Irrigator Center Pivot Energy Efficiency Test. Oklahoma Cooperative Extension Service, BAE-1411.
5. **Datta, S.**, Taghvaeian, S., & Stivers, J. (2017). Understanding Soil Water Content and Thresholds for Irrigation Management. Oklahoma Cooperative Extension Service, BAE-1537, Stillwater, OK, USA.

Conference Presentations – Oral

1. Pandit, A., **Datta, S.**, Taghvaeian, S., Kennedy, T., Bonham, R., Kettner, J., Mirchi, A., & Alian, S. (2026). Changes in Consumptive Water Use in Furrow and Subsurface Drip Irrigated Fields. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
2. Kalayambakam Janardhan, A. Y., **Datta, S.**, Taghvaeian, S., Crick, C., & Kundu, P. K. (2026). Data-

driven Modeling of Multi-Depth Soil Moisture in Irrigated Fields. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.

3. Pandit, A., **Datta, S.**, & Taghvaeian, S. (2026). Seasonal Irrigation Estimation Framework Using OpenET. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
4. Kettner, J., **Datta, S.**, Kalayambakam Janardhan, A. Y., DeJonge, K., & Thorp, K. (2026). Effects of Weather Forecasts on Irrigation Scheduling. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
5. **Datta, S.** (2026). Groundwater Conservation in the Ogallala Aquifer: An Evaluation of Agricultural Water Management Strategies in Oklahoma. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
6. **Datta, S.** (2026). Paving the Path Forward to a National Scale Irrigation Scheduler by Leveraging pyfao56. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
7. Pandit, A., **Datta, S.**, Mirchi, A., & Alian, S. (2026). Can OpenET Accurately Delineate the Agricultural Field Boundaries? 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
8. Meadi, S., **Datta, S.**, Kundu, P. K., Mahan, J., Taghvaeian, S., & Kennedy, T. (2026). Humidity Constraints on Canopy Temperature-Based Irrigation Scheduling Across Climatic Gradients in Oklahoma Cotton. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
9. **Datta, S.** (2026). Data-driven Modeling of Canopy Temperature for Variably Irrigated Cotton. 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
10. Meadi, S., Kalayambakam Janardhan, A. Y., & **Datta, S.** (2026). Can We Accurately Estimate Canopy Temperature Without Adjusting Sensor Heights to Reduce Labor Requirements? 2026 ASABE Annual International Meeting, Indianapolis, Indiana, July 12-15, 2026.
11. Pandit, A., **Datta, S.**, Taghvaeian, S., & Meadi, S. (2026). Estimation of Seasonal Irrigation At Place of Use Using OpenET. 2026 UCOWR Water Conference, San Antonio, TX, June 8-10, 2026.
12. Pandit, A., **Datta, S.**, Mirchi, A., & Alian, S. (2026). Evaluation of OpenET's Accuracy in Delineating Agricultural Field Boundaries. 2026 ASABE OK Sectional Meeting, Norman, OK, April 10, 2026.
13. Mohite, D., Singh, H., Barbadillo, M. A. M., Sharma, S., Lambert, L. H., **Datta, S.**, & Lofton, J. (2025). Yield Estimation for Rotational and Continuous Cotton Corn Cropping Systems Under Limited Well Capacities Using DSSAT Model. CANVAS 2025, Salt Lake City, UT, November 9-12, 2025. Available here.
14. Mirchi, A., Arshad, A., Akbar, M. U., Sadler, J. M., **Datta, S.**, Basagaoglu, H., & AghaKouchak, A. (2025). Downscaled GRACE/GARCE-FO Revealed Rapid Groundwater Storage Decline Driven by Irrigation Expansion in the Upper Red River Basin, United States. 2025 AGU Annual Meeting, New Orleans, LA, December 15-19, 2025.
15. Kettner, J., **Datta, S.**, Kalayambakam Janardhan, A. Y. (2025). Analyzing the Impacts of Irrigation Schedules Using Historical Weather Forecasts in pyfao56. 2025 Oklahoma Governor's Water Conference, Norman, OK, December 3-4, 2025.
16. Kalayambakam Janardhan, A. Y., **Datta, S.**, Crick, C., Taghvaeian, S., Kundu, P. K. (2025). Data-driven Modeling of Soil Moisture at Irrigated Fields. 2025 Oklahoma Governor's Water Conference, Norman, OK, December 3-4, 2025.
17. Pandit, A., **Datta, S.**, Taghvaeian, S., Kennedy, T., Bonham, R., Kettner, J., Mirchi, A., & Alian, S. (2025). Investigating Changes in Consumptive Water Use in Furrow and Subsurface Drip Irrigated Fields. 2025 Oklahoma Governor's Water Conference, Norman, OK, December 3-4, 2025.
18. Pandit, A., **Datta, S.**, Taghvaeian, S. (2025). A Seasonal Irrigation Abstraction Framework to Strategize Water Allocation. 2025 UCOWR Annual Meeting, Minneapolis, MN, June 3-6, 2025.
19. **Datta, S.** (2024). Leveraging Data: Scheduling Irrigations with Minimal Needs for Instrumentation. 2024 ASABE OK Sectional Meeting, Stillwater, OK, March 8, 2024.
20. **Datta, S.**, Taghvaeian, S., Sibley, M., Long, M., Gholson, D., Yost, M., Allen, N., Bali, Khaled, &

Zaccaria, D. (2023). Perception of US Farmers Towards Water Conservation in Surface Irrigation. 2023 ASABE International Annual Meeting, Omaha, Nebraska, July 9-12, 2023.

21. Islam, M. T., Munmun, T. H., Rahman, M. M., Adham, A.K.M., & **Datta, S.** (2023). Investigating the Effects of Raised Bed Width on Rice Cultivation in Bangladesh: Water Use and Yield. 2023 ASABE International Annual Meeting, Omaha, Nebraska, July 9-12, 2023.
22. Mehata, M., Taghvaeian, S., & **Datta, S.** (2022). Effect of Soil Data Accuracy on Irrigation Requirements of Major Crops in Oklahoma Using a Soil Water Balance Model. 2022 ASABE International Annual Meeting, Houston, Texas, July 17-20, 2022.
23. **Datta, S.** & Taghvaeian, S. (2022). Soil Water Sensors for Irrigation Scheduling in the United States: A Systematic Review of Literature. 2022 ASABE International Annual Meeting, Houston, Texas, July 17-20, 2022.
24. Mehata, M., Taghvaeian, S., **Datta, S.**, Bonham, R., & Ochsner, T. (2022). Performance Assessment and Application of Soil Water Sensors Under Variable Field Conditions. 2022 ASABE International Annual Meeting, Houston, Texas, July 17-20, 2022.
25. Taghvaeian, S., Mehata, M., **Datta, S.**, & Bonham, R. (2022). Estimating Water use of Irrigated Cotton Based on the Soil Water Content Data. 2022 ASCE EWRI World Environmental & Water Congress, Atlanta, Georgia, June 5-8, 2022.
26. Mehata, M., Taghvaeian, S., **Datta, S.**, & Moriasi, D. (2019). Modeling Soil Moisture Dynamics to Improve Irrigation Management. 2019 University of Oklahoma International WaTer Conference, Norman, Oklahoma, September 16-17, 2019.
27. **Datta, S.**, Taghvaeian, S., Mehata, M., & Moriasi, D. (2019). Irrigation Water Fluxes in an Agricultural Watershed in Central Oklahoma. 2019 ASABE Annual International Meeting, Boston, Massachusetts, July 7-10, 2019.
28. **Datta, S.**, Taghvaeian, S., Mehata, M., & Moriasi, D. (2018). Irrigation Water Fluxes in Fort Cobb Reservoir Experimental Watershed. 2018 39th Oklahoma Governor's Water Conference, Midwest City, Oklahoma, December 5-6, 2018.
29. Mehata, M., Taghvaeian, S., **Datta, S.**, & Moriasi, D. (2018). Use of a Computer Model to Simulate Soil Water Content in Irrigated Fields. 2018 39th Oklahoma Governor's Water Conference, Midwest City, Oklahoma, December 5-6, 2018.
30. Taghvaeian, S., Boman, R., Masasi, B., **Datta, S.**, & Stivers, J. (2018). Impact of Irrigation Termination Date and Soil Moisture on Cotton Yield. 2018 ASABE Annual International Meeting, Detroit, Michigan, July 29 – August 1, 2018.
31. **Datta, S.**, & Taghvaeian, S. (2018). Performance of Soil Moisture Sensors Under Field Conditions. 2018 ASABE Annual International Meeting, Detroit, Michigan, July 29 – August 1, 2018.
32. **Datta, S.**, Taghvaeian, S., Stivers, J., Ochsner, T., & Moriasi, D. (2017). Performance Evaluation of Soil Moisture Sensors Under Field Conditions. 2017 38th Oklahoma Governor's Water Conference, Norman, Oklahoma, October 31 – November 1, 2017.
33. **Datta, S.**, Rahman, S., Borhan, M.S., Saini-Eidukat, B., Cihacek, L., & Ringwall, K. (2016). Quantification and Characterization of Particulate Matter Generated from Unpaved Roads in the Oil Development Area of Western North Dakota. 2016 ASABE Annual International Meeting, Orlando, Florida, July 17-20, 2016.

Conference Presentations – Poster

1. Kundu, P. K., **Datta, S.**, Taghvaeian, S., Kennedy, T., & Sadler, J. M. (2026). Data-driven Modeling of Canopy Temperature Under Variably Irrigated Cotton. 2026 OSU Undergraduate Research Symposium and Wentz Scholars Presentation, Stillwater, OK, April 17 & 21, 2026.
2. Kennedy, T., Mirchi, A., Akbar, M. U., Khodkar, K., Alian, S., **Datta, S.**, Taghvaeian, S., & Arshad, A. (2025). A Spatial Water Balance Approach to Evaluate Groundwater Withdrawal and Irrigation Sus-

tainability in the Upper Red River Basin, US. 2025 AGU Annual Meeting, New Orleans, LA, December 15-19, 2025.

3. Kennedy, T., & **Datta, S.** (2025). Lake Lugert-Altus: Sustaining Irrigated Agriculture in Southwestern Oklahoma. 2025 OSU Undergraduate Research Symposium, Stillwater, OK, April 22, 2025.
4. Janardhan, A. Y. K., & **Datta, S.** (2025). Oklahoma Irrigation Scheduler: Paving the Path for Sustainable Water Management. 2025 ASABE OK Sectional Meeting, Stillwater, OK, March 7, 2025.
5. Kettner, J., Janardhan, A. Y. K., & **Datta, S.** (2025). Accuracy of Irrigation Scheduling Recommendations from Historical Weather Forecasts. 2025 ASABE OK Sectional Meeting, Stillwater, OK, March 7, 2025.
6. Pandit, A., **Datta, S.** (2025). Seasonal Irrigation Estimation Using Satellite Products. 2025 ASABE OK Sectional Meeting, Stillwater, OK, March 7, 2025.
7. Janardhan, A. Y. K., **Datta, S.** (2024). Oklahoma Irrigation Scheduler: Paving the Path for Sustainable Water Management. 2024 Oklahoma Governor's Water Conference, Norman, OK, November 19-20, 2024.
8. Kundu, P. K., **Datta, S.**, & Sadler, J. (2024). Canopy Temperature Prediction using State-of-the-Art Machine Learning Approaches for Irrigation Management of Cotton. 2024 Oklahoma Governor's Water Conference, Norman, OK, November 19-20, 2024.
9. Pandit, A., **Datta, S.**, Ochsner, T. E., & Wagle, P. (2024). Comparison of Field-Scale Evapotranspiration Estimates Based on In-situ Soil Moisture Sensors and Remote-sensing Approaches. 2024 Oklahoma Governor's Water Conference, Norman, OK, November 19-20, 2024.
10. Quellar, G., Kundu, P. K., & **Datta, S.** (2024). Mapping Crop Water Use in Lugert-Altus Irrigation District. 2024 ASABE OK Sectional Meeting, Stillwater, OK, March 8, 2024.
11. Kennedy, T., Pandit, A., & **Datta, S.** (2024). Impacts of Changing Weather Patterns on Irrigation Water Demand in Oklahoma. 2024 ASABE OK Sectional Meeting, Stillwater, OK, March 8, 2024.
12. Kundu, P. K., Pandit, A., & **Datta, S.** (2024). Machine Learning to Estimate Plant Canopy Temperature for Effective Irrigation Scheduling. 2024 ASABE OK Sectional Meeting, Stillwater, OK, March 8, 2024.
13. Pandit, A., Kundu, P. K., & **Datta, S.** (2024). Advancements in Soil Moisture Prediction Through Machine Learning Models in Irrigated Fields. 2024 ASABE OK Sectional Meeting, Stillwater, OK, March 8, 2024.
14. Bonham, R., Mehata, M., **Datta, S.**, & Taghvaeian, S. (2022). Effectiveness of Soil Moisture Sensors to Improve Irrigation Management. 2022 Research Day at the Capitol, (Virtually held), March 3, 2022.
15. Bonham, R., Taghvaeian, S., & **Datta, S.** (2021). Comparing Consumptive Water Use of Furrow and Drip Irrigated Cotton Fields. 2021 41st Oklahoma Governor's Water Conference and Research Symposium, (Virtually held), December 1-2, 2021.
16. Mehata, M., Taghvaeian, S., **Datta, S.**, Bonham, R., & Ochsner, T. (2021). Evaluating the Accuracy of Soil Moisture Sensors and Their Application for Irrigation Management. 2021 41st Oklahoma Governor's Water Conference and Research Symposium, (Virtually held), December 1-2, 2021.
17. Ochsner, T. E., Meek, A., Muller, L., Weckler, P., **Datta, S.**, Taghvaeian, S., Sibley, M., & Liesel, R. (2020). Use of Soil Moisture Data in Agricultural Water Management: On-farm Sensor Research and Perspectives from Irrigators. 2020 AGU Fall Meeting, (Virtually held), December 1-17, 2020.
18. Meek, A., Muller, L., Ochsner, T., Taghvaeian, S., Weckler, P., & **Datta, S.** (2020). Wireless Soil Moisture Sensor Performance in Irrigated Cropland: On-Farm Evaluation. 2020 ASA, CSSA and SSSA International Annual Meeting, (Virtually held), November 9-13, 2020.
19. **Datta, S.**, & Taghvaeian, S. (2020). A Critical Review of Soil Moisture Sensor Applications in Irrigation Management. 2020 ASABE OK Sectional Meeting, Stillwater, OK, February 20, 2020.
20. Alghamdi, R, **Datta, S.**, Cihacek, L, & Day, S. (2019). Changes in Crop and Soil Management Patterns on High Wind Erosion Risk Soils over 2 Decades. 2019 ASA, CSSA, and SSSA International Annual

Meeting, San Antonio, Texas, November 10-13, 2019.

21. Mehata, M., Taghvaeian, S., **Datta, S.**, & Moriasi, D. (2019). Simulating Soil Moisture Fluctuations under Irrigated Crops Using HYDRUS Model. 2019 ASABE Annual International Meeting, Boston, Massachusetts, July 7-10, 2019.
22. **Datta, S.**, & Taghvaeian, S. (2018). Performance of Soil Moisture Sensors Under Field Conditions. 2018 ASABE OK Sectional Meeting, Stillwater, OK, February 23, 2018.

Extension Presentations & Invited Talks

1. **Datta, S.** (2026). Oklahoma Agricultural Scientific Irrigation Scheduling – OASIS – Tool: A National Blueprint Towards Sustainable Irrigation Management. 2026 National Master Irrigator Summit, August 10-12, 2026, Lone Wolf, OK (50+ participants).
2. **Datta, S.** (2026). How do we use ET? 2026 OU ET Meeting on Vegetation Water Use and Stress Modeling Work in the South-Central US, University of Oklahoma, July 28, 2026, Norman, OK (12 participants).
3. **Datta, S.** (2026). Canopy Sensors, Telemetry, and AI: What's Actually Useful for Irrigators?. 2026 Minnesota Master Irrigator Program, March 10, 2026, Perham, MN & Virtual (18 participants).
4. **Datta, S.** (2026). Sustainable, Smart, Field-scale Agricultural Water Management Using AI-driven Sensor & Satellite Data Assimilation. 2026 OSU Civil & Environmental Engineering EWR Graduate Seminar, February 27, 2026, Stillwater, OK (20 participants).
5. **Datta, S.** (2025). Introduction to Oklahoma Agricultural Scientific Irrigation Scheduler - OASIS tool. 2025 OSU Winter CCA Crops School, December 16, 2025, Stillwater, OK (135 participants).
6. **Datta, S.** (2025). Canopy Temperature Modeling for Cotton Irrigation Management. OSU Cotton Field Day, September 25, 2025, Altus, OK (~35 participants)
7. **Datta, S.** (2025). Oklahoma Irrigation Scheduler. 2025 Oklahoma Master Irrigator Program, January 29, 2025. (19 participants)
8. **Datta, S.** (2025). Soil Water Relationships and Irrigation Scheduling. 2025 Oklahoma Master Irrigator Program, January 22, 2025. (17 participants)
9. **Datta, S.** (2024). Soil Water Relationships. 2024 Oklahoma Commissioners of the Land Office REMD Meeting, August 2, 2024. (15 participants)
10. **Datta, S.** (2024). Irrigation Research Overview and Potential Needs. OSU Field Research and Service Unit In-Service Training, Stillwater, OK, March 12, 2024. (~20 participants).
11. **Datta, S.** (2024). Soil-water Relationships. 2024 Oklahoma Master Irrigator Program, Woodward, OK, January 30, 2024. (25+ participants)
12. **Datta, S.** (2022). Irrigation Water Management in Maine: Why Now and How? Agricultural Water Management Board Meeting, Department of Agriculture, Conservation, and Forestry, (virtually held), October 27, 2022 (15 participants)
13. **Datta, S.** (2022). Engineered Tools & Approaches for Improved Wild Blueberry Production in Maine. Wild Blueberry Fresh Pack Line Meeting, October 11, 2022. (10 participants)
14. **Datta, S.** (2022). Webinar: Irrigation for Small-Scale Farmers: What do You Need to Know to Install a New Irrigation System? UMaine Cooperative Extension, (virtually held), August 22, 2022. (21 participants)
15. **Datta, S.** (2022). Importance of Irrigation for Wild Blueberry Producers in Maine. 2022 Blueberry Hill Farm Field Day, Jonesboro, ME, July 14, 2022. (80+ participants)
16. **Datta, S.** (2022). Integration of Engineered Approaches to Improve Profitability of Wild Blueberry Production in Maine. 2022 Wild Blueberry Commission of Maine Mid-Year Meeting, June 22, 2022. (15+ participants)
17. **Datta, S.** (2022). Battle Climate Change: How Irrigation Can Help Maine Vegetable and Fruit Growers. 2022 Maine Vegetable and Small Fruit Growers' Association Twilight Meeting, Auburn, ME, June 7,

2022. (35+ participants)

Student Supervision Activities

Students Directly Advised

Current Students

1. Jacob Kettner
   Ph.D. Candidate in BAE, Graduate Research/Teaching Assistant, Fall 2024 – 2028.
   Dissertation topic: Sensitivity analysis of input parameters such as soil data, weather data – current and forecasted and crop specific data on irrigation scheduling for major crops in Southern Great Plains.
2. Alok Pandit
   Ph.D. student in BAE, Graduate Research Assistant, Fall 2026 – Spring 2030.
   Thesis topic: Applying Q-Stable for Sustainable Groundwater Use in Oklahoma.
3. Abinava Yeshwanth Kalayambakam Janardhan
   Ph.D. student in BAE, Graduate Research Assistant, Spring 2026 – Fall 2029.
   Dissertation topic: Application of AI in soil moisture-based irrigation management.
4. Satheesh Meadi
   Ph.D. student in BAE, Graduate Research Assistant, Spring 2026 – Fall 2029.
   Dissertation topic: Development of data-driven, sensor-enhanced machine/deep learning prediction algorithm of canopy temperature for irrigated cotton.

Past Students

1. Alok Pandit
   M.S. in BAE, Graduate Research Assistant, Spring 2024 – Summer 2026.
   Thesis topic: Feasibility analysis of using OpenET for irrigation water management. Status: Joined my team as a Ph.D. student starting August, 2026.
2. Tatumn Kennedy
   B.S. in BAE, Undergraduate Research Assistant, Spring 2024 – Spring 2026.
   Research topic: Quantifying crop water use among major irrigation systems in Southwest Oklahoma and identification of relative humidity limited areas for implementation of canopy temperature-based irrigation scheduling algorithms.
   Status: Graduated and joined OSU BAE as a M.S. student.
3. Abinava Yeshwanth Kalayambakam Janardhan
   M.S. student in Computer Science, Graduate Research Assistant in my teem, Summer 2024 – Fall 2025.
   Thesis topic: Data-driven Modeling of Soil Moisture in Irrigated Croplands.
   Status: Joined my team at OSU BAE as a Ph.D. Student.
4. Palash Kumar Kundu
   M.S. student in BAE and Graduate Research Assistant. Spring 2024 – Summer 2025.
   Thesis topic: Data-driven modeling of canopy temperature for irrigated cotton.
   Status: Graduated and pursuing Ph.D. at Michigan State University.
5. Gillian Quellar
   B.S. student in BAE, Undergraduate Research Assistant, Spring 2024.
   Research topic: Quantifying ET in Southwest Oklahoma using OpenET platform.

As Graduate Committee Member

Current Students

1. Masrufa Bayesh, Ph.D. Student, Computer Science. Anticipated graduation: Fall 2028
2. Naiem Sheikh, Ph.D. Student, Geography. Anticipated graduation: Fall 2028.

3. Digvijay Mohite, Ph.D. Student, Plant and Soil Science. Anticipated graduation: Fall 2026.
4. Kasra Khodkar, Ph.D. Student, Biosystems & Agricultural Engineering. Anticipated graduation: Spring 2026.
5. Muhammad Umar Akbar, Ph.D. Student, Biosystems & Agricultural Engineering. Anticipated graduation: Spring 2026.
6. Navdeep Kaur Saasan, Ph.D. Student, Biosystems & Agricultural Engineering. Anticipated graduation: Spring 2026.

Past Students

1. Cole Diggins, Ph.D. Student, Soil Science. Graduated: Summer 2026.
2. Mark Barbadillo, Ph.D., Soil Science. Graduated: Spring 2026.
3. Mamata Pandey, M.S., Biosystems & Agricultural Engineering. Graduated: Summer 2025.
4. Jacob Kettner, M.S., Biosystems & Agricultural Engineering. Graduated: Spring 2024.

Teaching Activities

Overview

I aim to foster an engaging and inclusive learning environment that inspires students to think critically and solve problems and develop a deeper understanding of their field of study. Applying a combination of flipped classroom, hands-on activities, and interactive lectures bringing theoretical knowledge with real-word applications, I strive to cultivate curiosity, creativity, and collaboration among students. I teach two major courses within Biosystems and Agricultural Engineering - BAE and Agricultural Systems Technology - AST curriculums at OSU and participates in various student recruitment activities.

Taught Courses

1. **AST 4203, 5200: Agricultural Water Management**

Term	UG[1]	G[2]	Mean Course Evaluation Scores		
			Enhanced Learning[3]	Overall Course Rating[4]	Overall Instructor Rating[5]
Fall 2023	14	6	N/a	N/a	3.17[6]
Fall 2024	12	3	3.20	3.20	3.60
Fall 2025	11	0	3.67	2.89	3.33

[1] Number of Undergraduate Students.
[2] Number of Graduate Students.
[3] Mean response to the statement: "Presentation of course content enhanced my learning of the subject"; Scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree.
[4] Mean response to the statement: "I would describe this as an excellent course"; Scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree.
[5] Mean response to the statement: "I would describe this instructor as an excellent teacher"; Scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree.
[6] Did not receive enough responses (only 2) in OSU Student Survey of Instruction (OSU SSI). I conducted my own survey through Qualtrics. By December 12, 11 students completed the survey. Therefore, the rating is not official and only the mean score for the last statement is provided.

Short summary: In Fall 2023, I co-taught AST 4203/5200 with Dr. Paul Weckler, now retired. The lecture-only course met twice weekly for 75 minutes. Student feedback noted: (1) limited hands-on activities

made concepts such as distribution uniformity and soil moisture measurement difficult to visualize; (2) more guided mathematical problem-solving was needed before homework; and (3) infrequent Canvas grade updates made progress tracking difficult.

In Fall 2024, I redesigned the course based on this feedback by: (1) adding a weekly 1 hour 50 minute lab and shortening lectures to two 50-minute sessions; (2) developing six hands-on labs with BAE Ph.D. student and TA Jacob Kettner on soil moisture measurement, mapping, distribution uniformity, and related topics; (3) adding a research tour to the Cimarron Valley Research Station in Perkins, OK; (4) inviting five faculty guest lecturers on soil moisture, irrigation systems, water quality, flood analysis, salinity, and drainage; (5) adding mathematical workout sessions before homework; (6) updating Canvas grades weekly; and (7) revising grading to emphasize homework and in-class quizzes over exams. Student performance and feedback suggested Fall 2025 improvements: (1) reduce guest lectures; (2) add end-of-class quizzes; and (3) move lectures from BAE Lab Building Room 129 due to air-conditioning noise.

In Fall 2025, I: (1) moved lectures to Agricultural Hall 342; (2) reduced guest lectures to one targeted lecture by Dr. Tyson Ochsner on soil moisture measurement; and (3) revised grading to emphasize frequent end-of-class pop-up quizzes, in addition to announced quizzes. These quizzes, developed by Jacob Kettner from class topics, promoted attendance and engagement. Fall 2026 improvements will include: (1) posting in-class math examples on Canvas before class; and (2) seeking a later class time.

2. **BAE 5030: Advanced Techniques in Irrigation Water Management**

Term	G	Mean Course Evaluation Scores		
		Enhanced Learning	Overall Course Rating	Overall Instructor Rating
Spring 2025	5	4.50	4.25	4.00

Short summary: I designed this course from scratch and taught it in Spring 2025 with curriculum on advanced techniques on irrigation water management such as soil moisture sensors, canopy temperature sensors, soil water balance models, and crop models. The course was ”Lecture” only taught for 2 hours 50 mins on each Thursday. Some observations based on student feedback were: (1) students liked the Extension Factsheet development and indicated that they gained deeper appreciation for the subject; and (2) completion of all contents outlined in syllabus and organization of course contents were recommended. Based on the feedback, planned future modifications for Spring 2027 were as follows: (1) requested change in class date and format to Monday and Wednesday having two Lecture sessions and one laboratory session on Wednesday, to have more discrete interactions with the students; (2) workload balanced and organized homework created beforehand will be handed out and mentioned in syllabus; and (3) efforts will be made to balance the course contents and to make sure that all contents are covered throughout the semester.

3. **SOIL 5583: Soil Physics Measuring Techniques**

Short summary: As Graduate Teaching Assistant (GTA) in Spring 2019, I taught this course to complete Ph.D. degree requirement with Instructor Dr. Tyson Ochsner in Plant and Soil Sciences Department. The course had 3 graduate students. As GTA, I developed two completely new laboratory exercises on soil electrical conductivity measurements, along with supervising other previously developed laboratory exercises. Rating as GTA is not available through OSU SSI (official survey).

Guest Lectures

1. **Datta, S.** Fall 2025. Delivered a 1-hour 50-minute combined lecture-laboratory session on ”Importance of Irrigation Research” and ”Soil Moisture Mapping” in the course BAE 1011 – Introduction to Biosystems Engineering. ~12 BAE students per session. 2 sessions. November 19-20, 2025.
2. **Datta, S.** Fall 2025. Delivered a 1-hour 15-minute lecture on ”Understanding Soil Moisture Data for

Irrigation Research" in the course BAE 6520 Advanced Water Resources Management. October 28, 2025.

3. **Datta, S.** Fall 2024. Delivered a 1-hour 50-minute combined lecture-laboratory session on "Importance of Irrigation Research" and "Soil Moisture Mapping" in the course BAE 1011 – Introduction to Biosystems Engineering. ~12 BAE students per session. 2 sessions. December 3 & 4, 2024.
4. **Datta, S.** Fall 2024. Delivered a 50-min lecture on "Irrigation Water Quality and Impacts of Marginal Water on Crop Yield" in the course BAE 4324 – Water Quality Engineering. ~12 BAE students. November 12, 2024.
5. **Datta, S.** Spring 2024. Delivered a 1-hour 50-minute laboratory session on "Measuring temperature and controlling LEDs using Campbell Scientific Dataloggers" in AST 4303 – Sensors and Controls for Agriculture. ~12 AST students. February 7, 2024.
6. **Datta, S.** Fall 2021. Delivered two 1-hour 15-minute lecture sessions on "Irrigation Principles and Systems" in the course AST 4203 – Agricultural Water Management. ~12 AST students. October 19 & 21, 2021.
7. **Datta, S.** Fall 2021. Delivered one 50-minute lecture session on "Design Considerations for Irrigation Systems" in the course BAE 4314 – Design Hydrology. ~12 BAE students. November 2 & 4, 2021.
8. **Datta, S.** Fall 2017. Delivered one 2-hour 50-minute lecture session on "Irrigated vs Rainfed Farming – Food-Water Tradeoff in Semi-arid Watershed." in the course NREM 5483 – Ecohydrology. ~10 students from different disciplines. October 23, 2017.

Student Recruitment Events

At Oklahoma State University

1. **Datta, S.**, & Kettner, J. (2025). Demonstration on Soil Moisture Mapping and its Impact on Irrigation Water Management. OSU College of Engineering, Architecture, and Technology (CEAT) Summer Bridge Engineering Student Recruitment Program. ~23 participants. July 30, 2025.
2. Mirchi, A., & **Datta, S.** (2023). Demonstration of Streambank Erosion through BAE Stream Trailer. OSU College of Engineering, Architecture, and Technology (CEAT) Summer Bridge Engineering Student Recruitment Program. ~15 participants. August 4, 2023.
3. Sadler, A., Khodkar, K., & **Datta, S.** (2023). Demonstration of Soil Moisture Sensing. OSU CEAT Summer Bridge Engineering Student Recruitment Program. ~10 students, August 4, 2023.
4. Samimi, M., **Datta, S.**, Mirchi, A., &Taghvaeian, S. (2021). Demonstration of Streambank Erosion through BAE Stream Trailer. OSU CEAT Summer Bridge, Discovery Day, and Upward Bound programs. ~20 students. July 30, August 2, 2021.
5. Taghvaeian, S., & **Datta, S.** (2019). Demonstration of Soil Moisture Mapping. OSU CEAT Summer Bridge Engineering Student Recruitment Program. ~7 students. July 31, 2019.

At North Dakota State University

1. **Datta, S.**, Borhan, M. S., & Rahman, S. (2015). Demonstration of Air Quality Monitoring Techniques. North Dakota State University-Nueta Hidatsa Sahnish College Combined Air Quality Monitoring Program. 3 students. ~October, 2015.

Tools Developed

Operational Agriculture and Soil Information System – OASIS Tool

Status: Already developed and released to beta-testers on May 15, 2026.

Address: https://irrigation.okstate.edu

Short summary: Operational Agriculture and Soil Information System – OASIS – previously known as

Oklahoma Agricultural Scientific Irrigation Scheduler, is a the irrigation scheduling tool, built from ground up, for agricultural irrigators in Oklahoma, but with versatility to be applied anywhere within the U.S. This tool is built on dual-crop coefficient-based soil water balance model that can provide more accurate estimate of crop water use – evapotranspiration (ET) and let users schedule irrigations based on Oklahoma Mesonet weather data and in-field irrigation data. More improvements for this tool are on the horizon (e.g., Growing Degree Days based approach, soil water data infusion, OpenET integration) and will be implemented in near future. The tool is also being expanded to Kansas and Texas. This tool's development and deployment has attracted over **$2M** as of October 16, 2026.

Media Mentions & Public Engagement

1. *OSU Agriculture programs help Oklahomans protect water resources.* Jemella Zhang. August 24, 2026. Quoted on satellite evapotranspiration and regional water accounting; OASIS named. OSU Extension, extension.okstate.edu; reprinted by OKW News.
2. *2026 Master Irrigator Summit Convened 50 Water Leaders from 15 States.* August 19, 2026 (posted August 21). Named as presenter of the OASIS tools-and-data demonstration. Irrigation Innovation Consortium, Colorado State University, irrigationinnovation.org.
3. *Kennedy named 2026 Ferguson College of Agriculture Outstanding Senior.* Dean Ruhl. March 27, 2026. Named as research supervisor of the awardee, on cotton canopy temperature and irrigation scheduling. OSU News, news.okstate.edu; reprinted by Oklahoma Farm Report, March 30, 2026.
4. *OSU assists in the search for sustainable irrigation solutions.* Dean Ruhl, Communications & Media Relations Specialist. November 13, 2025. Named as head of OSU's OpenET work on the SatRAP project and quoted twice on field-scale evapotranspiration. OSU Ag Research, agresearch.okstate.edu; reprinted by Oklahoma Farm Report (Nov. 13) and Farm Talk News (Nov. 18).
5. *Introduction to Oklahoma Agricultural Scientific Irrigation Scheduler – OASIS Tool.* Presenter. 2025. Invited presenter, Day 1 Session 1; session recording posted publicly. Winter CCA Crops School, wintercropsschool.com.
6. *Irrigation.* Contributor, with Sumit Sharma and Kevin Wagner. September 2025. Named contributor on sensor-based irrigation scheduling. *Down and Dirty with NPK*, OSU Plant and Soil Sciences, osunpk.com.
7. *OSU Agriculture offers research, education to extend life of Ogallala Aquifer.* Gail Ellis, Editorial Communications Coordinator. May 8, 2025. Profiled and quoted twice on field-scale water accounting and producer adoption. OSU Extension, extension.okstate.edu; reprinted by Oklahoma Farm Report (May 9), High Plains Journal (May 14), Farm Talk News (May 20), and Morning Ag Clips (May 23).
8. *Data-driven Modeling of Remotely Sensed Canopy Temperature for Improved Irrigation Scheduling.* Faculty mentor. 2023-2024. Named as faculty mentor on a Wentz Research Scholar project. OSU Academic Affairs, academicaffairs.okstate.edu.
9. *Irrigation Scheduling for Potato in Maine.* Presenter. March 2023. Invited speaker, 2023 Sustainable Agriculture Workshop. University of Maine Cooperative Extension, extension.umaine.edu.
10. *Irrigation for Small-Scale Farmers.* Presenter. August 22, 2022. Sole presenter; recording posted publicly. University of Maine Cooperative Extension, extension.umaine.edu.
11. *UMaine Extension hires agricultural engineer Sumon Datta.* No byline. June 23, 2022. Subject of the article, on appointment as Assistant Extension Professor and Extension Agricultural Engineer. UMaine News, umaine.edu; reprinted by Bangor Daily News (June 24), Morning Ag Clips (June 27), and the *Portland Press Herald*.

Service & Recognition

Proposal Reviewer

- 2023 University of Maine PFAS Pilot Research Fund

Journal Manuscript Reviewer

- Journal of the ASABE
- MDPI: Sensors
- MDPI: Water
- MDPI: Agriculture
- Springer Nature: Irrigation Science
- Agricultural Water Management
- Environmental Research Letters
- Biosystems Engineering
- ASCE: Journal of Irrigation and Drainage Engineering
- Resources, Environment, and Sustainability
- Journal of Natural Resources and Agricultural Ecosystems
- Applied Engineering in Agriculture

Editorial Boards

Associate Editor 2024 – Present
Natural Resources and Environmental Systems, American Society of Agricultural and Biological Engineers (ASABE)

Professional Memberships

- American Society of Agricultural and Biological Engineers (ASABE) 2016 – Present
- Alpha Epsilon Honor Society 2016 – Present
- Institute of Engineers, Bangladesh 2014 – Present

Committees

- ASABE Committees
  - NRES-24 Irrigation Group 2022 – Present
  - NRES-244 Irrigation Management 2025 – Present
    - Elected as Vice Chair
  - ASABE OK Sectional Meeting 2025 – Present
    - Treasurer, 2025-2026
    - Vice-chair, 2026-2027
- ASCE Evapotranspiration, Irrigation, and Hydrology Task Committee 2020 – Present
- WERA-1022: Irrigation Scheduling Technologies Multi-State Hatch Project 2023 – Present
- W5128: Micro-irrigation Multi-State Hatch Project 2023 – Present
- University Committees
  - Search Committee
    - Extension State Program Specialist in Mesonet Agriculture Outreach 2025 & 2026
    - Soil & Water Conservation Extension Specialist, Plant & Soil Sciences 2024
  - Departmental Committee
    - BAE Graduate Committee 2023 – Present
    - BAE Awards Committee 2023 – 2024
    - BAE Research and Technology Development Committee 2023 – Present
    - BAE RPT&CR Committee 2024 – 2025
    - BAE Digital Presence & Communication Committee 2024 – Present

Awards

Student Awards

1. PhD student, Jacob Kettner, was awarded the Kathy & Leo Noltsenmeyer Graduate Fellowship to advance his PhD research in precision irrigation management in March, 2026.
2. Undergraduate student, Tatumn Kennedy, was named the Outstanding Senior in Ferguson College of Agriculture and the outstanding BAE sensior in College of Engineering, Architechture, and Technology at OSU in for Spring 2026.
3. Undergraduate student, Tatumn Kennedy, won the 2025 Oklahoma Governor's Water Conference Poster Presentation on December 3, 2025.
4. Undergraduate student, Tatumn Kennedy, received prestigious Wentz Scholarship at OSU to advance data-driven modeling of canopy temperature for cotton from Fall 2025 to Spring 2026.
5. Undergraduate student, Tatumn Kennedy, won the first prize for poster presentation on "Impacts of Changing Weather Patterns on Irrigation Water Demand in Oklahoma" at 2024 ASABE OK Sectional Meeting.
6. Undergraduate Student, Rio Bonham, won the first prize on poster presentation on "Comparing Consumptive Water Use of Furrow and Drip Irrigated Cotton Fields" at 2021 Oklahoma Governor's Water Conference.

My Awards

1. 2020 Outstanding Graduate Student at Oklahoma State University College of Engineering, Architecture, and Technology for maintaining a foundation of academic excellence and leadership.
2. 2020 DGF for Academic Excellence at Oklahoma State University.
3. 3rd Place in 3MT (3-minute-thesis) competition in 2020 ASABE OK Sectional Meeting.
4. 2019 Irrigation Association E3 Learner Education and Travel Award.
5. 2019 DGF for Academic Excellence at Oklahoma State University.
6. 3rd Place in 3MT (3-minute-thesis) competition in 2019 ASABE OK Sectional Meeting.
7. 2018 American Society of Agricultural and Biological Engineers (ASABE) 'Blue Ribbon Educational Award' for Extension Publication "Understanding Soil Water Content and Thresholds for Irrigation Management"
8. 2018 Distinguished Graduate Fellowship (DGF) for Academic Excellence at Oklahoma State University
9. Frank Bain Scholarship for Academic Excellence at North Dakota State University
10. Dean's List for Academic Excellence at Bangladesh Agricultural University
11. Dean's List for Academic Excellence at Bangladesh Agricultural University